December 2025 CSCA Chemistry

Atomic number H: 1 O: 16 Na: 23 Cu: 64

Select this question (there are 4 options per question, and only 1 is correct)

1. Which of the following changes is a chemical change? (...)

A. Iron rusts B. Activated carbon adsorption of pigments

C. volatilization of concentrated hydrochloric acid D. Ice melts into water.

Analysis: A The formation of new substances (iron oxide) through iron rusting is a chemical change.2.

The following compounds are clearly correct (...)

A.C0 Carbon dioxide B.$FeCO_3$ ferrous carbonate

C.Na_2CO_3 Baking soda D.CuCl – Copper chloride

Analysis: CO is carbon monoxide Na_2CO_3 is sodium carbonate CuCl is cuprous chloride

3. Which of the following compound names is incorrect?

A.$CH_3CH_2$0H Ethanol B.CH_2=CH_2 ethylene

C.HC Three HC Acetylene D.CH_3CH_3 Ethane Analysis: HC 三 CH Acetylene

4. Which of the following chemical terms is correct?

A.2 One hydrogen molecule: 2 H B.4 Water molecules:H_2O_2

C.5 sodium ion: $5Na^+$ D.3 Oxygen atoms 3O

5. Which of the following substances are pure substances?

A. hydrochloric acid B.Ice-water mixture

C. air D.Saturated saltwater

An ice-water mixture contains only water as a single substance (ice and water are two distinct states), thus it is classified as a pure substance.

6. A group of substances that are allotropes of each other is ()

A. White phosphorus and red phosphorus B. Ice and dry ice

C. Water and hydrogen peroxide D. Oxygen and liquid oxygen

Analysis: Isomorphs are different elements composed of the same element. White phosphorus and red phosphorus are composed of phosphorus and belong to isomorphs as different elements.

7. The oxidation number of N in NH_4NO_3 is ()

A. -3, +3 B. -3, +5

C.+1，+5　　D.+1，-5

Analysis:NH_4NO_3 can be split into NH_4^+ and NO_3^-

In NH_4, the oxidation state of H is +1. Let the oxidation state of N be a, then a+4 (-1) = +1, solving for a = -3. In N_0, the oxidation state of O is-2. Let the oxidation state of N be b, then b+3 (-2) = -1·, solving for b = +5.

8. Which of the following substances is a solution? (...)

A. distilled water B. normal saline

C. arachis oil D. cement paste

Analysis: Physiological saline is a homogeneous and stable mixture formed by the dissolution of sodium chloride in water, which belongs to the category of solutions.

9. When the flame color reaction is performed with sodium chloride, the flame appears (...).

A. pale asphyxia B. yellow

C. nattier blue D. green

Analysis: The flame color reaction of sodium produces a yellow flame.

10. Among the following substances, which one is a non-electrolyte?

A. natrium aceticum B. saccharose

C. caustic soda D. sulphuric acid

Analysis: Non-electrolytes are compounds that do not conduct electricity in both aqueous solution and molten states. Sodium acetate and sodium hydroxide are electrolytes, sulfuric acid is an electrolyte, while sucrose is classified as a non-electrolyte.

11. Which of the following substances is an electrolyte?

A.NaCl B. aluminium metallicum

C. saccharose D. carbon dioxide

Analysis: Electrolytes are compounds that can conduct electricity in aqueous solution or molten state. NaCl can conduct electricity in both aqueous solution and molten state, thus it is classified as an electrolyte; metallic aluminum is a pure element, neither electrolyte nor non-electrolyte; sucrose and carbon dioxide are non-electrolytes.

12. Which of the following reactions is an oxidation-reduction reaction?

A.$CaCO_3 \Delta CaO + CO_2\uparrow$ B.$HCl + NaOH \rightarrow NaCl + H_2O$

C.$Fe + CuSO_4 \rightarrow FeSO_4 + Cu$ D.$CO_2 + H_2O \rightarrow H_2CO_3$ Analysis:

The characteristic of redox reaction is the increase or decrease of the valence of elements.

A: The oxidation states of all elements remain unchanged, indicating a non-redox reaction.

B: Acid-base neutralization reaction, with no change in valence, is a non-redox reaction;

C: The oxidation state of Fe changes from 0 to +2, while that of Cu shifts from +2 to 0. This shift in oxidation states indicates a redox reaction.

D: The oxidation states of all elements remain unchanged, indicating a non-redox reaction.

13. Among the following substances, the one commonly used as an oxidizing agent is ().

A. iron B. carbon monoxide

C. mineral chameleon D. hydrogenation

Analysis: The oxidizing agent gains electrons in the reaction, resulting in a reduction in its oxidation state. Potassium permanganate (KMnO_4) contains manganese (Mn) in the +7 oxidation state, exhibiting strong oxidizing properties and is commonly used as an oxidizing agent. Iron, carbon monoxide, and sodium hydride are frequently employed as reducing agents.

14. Which of the following groups are allotropes of each other?

A. All-inclusive stone and stone within stone

B.CH_4 and C_4H_8

C. ^{12}C take part in ^{14}C

D.CO and CO_2

Analysis: Isomorphs are different elements composed of the same element. Diamond and graphite are both composed of carbon, which are different elements and are isomorphs of each other;CH_4 and C_4H_8 are compounds,^{12}C and ^{14}C are isotopes, and CO and CO_2 are compounds, which are not isomorphs.

15. Which of the following compounds has an acidic aqueous solution?

A.NaCl

B.NaOH

C.Na_2CO_3

D.NH_4Cl

Analysis:NH_4 Cl is a strong acid and weak base salt. NH_4^+ hydrolysis makes the aqueous solution acidic; NaCl solution is neutral, NaOH solution is alkaline, and Na_2CO_3 solution is alkaline because of CO_3^{2-} hydrolysis.

16. When phenolphthalein solution is added to the solution, if it turns red, the solution is ().

A. basicity

B. neutral

C. acidity

D. unidentified

Analysis: Phenolphthalein reagent turns red in alkaline solutions but remains unchanged in neutral or acidic solutions. Therefore, the red coloration indicates that the solution is alkaline.

17. Which of the following gases can be collected using both the drainage method and the upward air displacement method?

A. hydrogen

B. oxygen

C. carbon dioxide

D. methane

Analysis: A. Since hydrogen is less dense than air, it cannot be collected using the upward air displacement method. Therefore, option A is incorrect.

Since oxygen is denser than air, it can be collected by upward displacement. As oxygen is insoluble in water, it can be collected by displacement. Therefore, option B is correct.

C. Methane is less dense than air and cannot be collected using the upward air displacement method, therefore option C is incorrect.

D. Since carbon dioxide is water-soluble, it cannot be collected by drainage, making option D incorrect.

18. From 1L1mol. L^{-1} of NaOH Solution, Remove 50Ml of NaOH Solution, the 50ml, NaOH Solution of the Solution Concentration of ()

A.0.02$mol \cdot L^{-1}$

B.1 $mol \cdot L^{-1}$

C.0.2 $mol \cdot L^{-1}$

D.2 $mol \cdot L^{-1}$

Analysis: The solution is homogeneous. Take 50mL from 1L1mol. L^{-1} NaOH solution, the concentration of solute is still 1mol. L^{-1}.

19. Which of the following statements about the use of substances is incorrect?

A.02 assisted combustion B.CO can be used to put out fires.

C. Carbon can be used as fuel

D.75% alcohol can be used for disinfection

Analysis: CO is flammable and cannot be used to put out fires;O_2 it can be used to help combustion, carbon can be used as fuel, and 75% alcohol can be used for disinfection. All three statements are correct.

20.^{16} S position in the periodic table ()

A. Week 3, Group IVA B.The 3rd cycle, VIA group

C. Week 2, Group IVA D.The 2nd cycle, VA group

Analysis: Sulfur (S) has an atomic number of 16, with an electron configuration of 2, 8, 6, comprising three electron shells in the third period. Its outermost shell contains six electrons, placing it in Group VI A.

21. When a cup of concentrated sulfuric acid is left in the air, what changes occur after a period of time?

A. mass reduction of solvent

B.The mass of the solution has increased.

C. increase in the mass of solute

D.The mass of the solution has decreased.

Analysis: Concentrated sulfuric acid is hygroscopic, which will absorb water vapor in the air, resulting in the increase of the mass of the solvent in the solution, and the total mass of the solution will also increase; the mass of the solute H_2SO_4 remains unchanged.

22. Which of the following groups of compounds are homologues?

A.CH_3CH_3 and HC three CH B.$CH_3CH{=}CH_2$ and CH_3CH_3

C.$HC_3CH{=}CH_2$ and $CH_3C{=}CCH_3$ D.CH_3CH_3 and $CH_3CH_2CH_3$

Analysis: Homologues refer to organic compounds with similar structures and one or more CH_2 groups in their molecular composition.

D:CH_3CH_3 and $CH_3CH_2CH_3$ are both alkanes with similar structures and a difference of 1 CH_2 in molecular composition.

23. Which of the following reaction types is correctly described?

A.MnO_2+ 4HCl (concentrated) → $MnCl_2 + Cl_2\uparrow + 2H_2O$ Chemical reaction

B.Fe_2O_3+ 3CO → 2Fe +$3CO_2$ Substitution reaction

C.3Fe+$2O_2$ → Fe_2O_4 Decomposition Reaction

D.HClO + NaOH → NaClO +H_2O double decomposition reaction

analyze

A:MnO_2+ 4HCl (concentrated) → $MnCl_2 + Cl_2\uparrow + 2H_2O$, this reaction is not a combination reaction (combination reaction is the formation of one substance from many substances);

B:Fe_2O_3+ 3CO → 2Fe +$3CO_2$, there is no elemental substance in the reactants, so it is not a displacement reaction (displacement reaction is the reaction of elemental substance and compound to generate new elemental substance and new compound);

C: 3Fe +$2O_2$ → Fe_2O_4, this reaction is a combination reaction, not a decomposition reaction (decomposition reaction is a kind of substance to produce a variety of substances);

• D: HClO + NaOH → NaClO +H_2 O is a double decomposition reaction in which two compounds exchange components to form two other compounds.

24. When the reaction reaches equilibrium, the constant in the reaction system is ().

A. A. The atom exists in the product molecule of the reactant molecule.

B. The time of reactant concentration and product concentration.

C. The rate of the reverse reaction is the reciprocal of the forward reaction rate. D. Mass of reactants and mass of products

Analysis: When a chemical reaction reaches equilibrium, the forward reaction rate equals the reverse reaction rate (and is not zero), which is the core characteristic of chemical equilibrium.

A: Atoms are always present in the particles of reactants and products, not a phenomenon unique to equilibrium.

B: During equilibrium, the concentrations of reactants and products may not be equal, but they remain constant.

D: The mass of reactants and products may not be equal at equilibrium. Mass conservation is a universal law of chemical reactions, independent of whether equilibrium is reached.

25. Which of the following substances is an oxide?

A.H_2O B.NaOH

C.O_2 D.$KClO_3$

Analysis: The definition of an oxide is a compound composed of two elements, one of which is oxygen.

A:H_2 O is composed of hydrogen and oxygen and belongs to the oxide;

B: NaOH is composed of three elements—sodium, oxygen, and hydrogen. It is an alkali, not an oxide.

C:O_2 is a simple substance composed of oxygen, not a compound, and does not belong to the oxide;

D:$KClO_3$ is composed of potassium, chlorine and oxygen, and belongs to salt, not oxide.

26 Which of the following statements about catalysts is incorrect?

A. The chemical properties of the catalyst do not change before and after the reaction

B.The mass of the catalyst changes before and after the reaction

C. Certain reactions can utilize different catalysts. D. Catalysts can alter the reaction rate

Analysis: The characteristic of catalyst is "one change and two constants": it can change the reaction rate (can accelerate or slow down), and the mass and chemical properties remain unchanged before and after the reaction.

A: The chemical properties of the catalyst remain unchanged before and after the reaction, which is correct.

B: The statement that the catalyst's mass remains unchanged before and after the reaction is incorrect.

C: Some reactions can employ different catalysts (e.g., manganese dioxide or copper oxide can serve as catalysts for hydrogen peroxide decomposition), which is correct.

D: The statement that catalysts can alter reaction rates is correct.

27. Which of the following groups contains the same number of chemical bonds?

A.NaCl and HCl B.O_2 and H_2

C.HNO_3 and NH_4NO_3 D.NaOH and KCl

Analysis: Chemical bonds are classified into ionic bonds and covalent bonds (non-polar covalent bonds and polar covalent bonds). The following options are analyzed sequentially:

A: NaCl contains ionic bonds, while HCl has polar covalent bonds, indicating different bond types.

B:O_2 and H_2 are both elements, and they contain nonpolar covalent bonds in their molecules, with the same bond type;

C:HNO_3 contains only covalent bonds, while NH_4NO_3 contains ionic bonds and covalent bonds, with different bond types;

D: NaOH contains both ionic bonds and polar covalent bonds, whereas KCl contains only ionic bonds, indicating different bond types.

28. Which of the following glassware is called a beaker?

A. 

B. 



C.



D

analyze ：

A is a volumetric flask, used to prepare solutions with a specific molar concentration.

B is a standard funnel, primarily used for filtration operations;

C is a beaker, which can be used for dissolving, preparing solutions, heating liquids, etc.

D is a conical flask, commonly used for titration experiments and as a reaction vessel.

29. When $BaCl_2$ solution is added to a solution, a white precipitate is formed, and the precipitate does not dissolve after adding nitric acid.

A. The solution must contain $S024^{-}$

B.The solution may contain c023^{-}

C. The solution must contain Ag^{+}

D. The solution may contain S 024^{-}or Ag^{+}. Analysis: Add $BaCl_2$ to produce white precipitate, nitric acid does not dissolve, the precipitate may be $BaSO_4$ or Agcl.

-S024^{-}and Ba $^{2+}$form $BaSO_4$ precipitate, insoluble in nitric acid;

-Ag^{+}and cl-form AgCl precipitate, which is insoluble in nitric acid;

The $BaCO_3$ precipitate formed by c-023^{-}and Ba $^{2+}$is soluble in nitric acid, so the solution cannot contain c023^{-}.

30. The reaction between hydrochloric acid and sodium hydroxide is classified as ().

A. decomposition reaction　　B. neutrallzation reaction

C. combination reaction　　D. displacement reaction

Analysis: The reaction between hydrochloric acid (acid) and sodium hydroxide (base) produces sodium chloride (salt) and water. This type of reaction, where acids and bases form salts and water, is classified as a neutralization reaction.

31. Which of the following groups of elements belongs to the same group?

A. Hydrogen, Lithium, Sodium　　B.Carbon, oxygen, sulfur

C. Hydrogen, carbon, nitrogen　　D.Sodium, magnesium, aluminum

Analysis: In the periodic table of elements, the outermost electron count is identical for elements within the same group (column).

A: Hydrogen (H), lithium (Li), and sodium (Na) are all Group IIA elements, with 1 valence electron in their outermost shell.

B: Carbon (C, IVa), oxygen (O, VIa), and sulfur (S, VIa) do not belong to the same group.

C: Hydrogen (ⅠA), carbon (ⅣA), and nitrogen (ⅤA) do not belong to the same group;

D: Sodium (ⅠA), magnesium (ⅡA), and aluminum (ⅢA) belong to the same period, not the same group.

32. Which of the following is the correct atomic radius gradient rule for allicin?

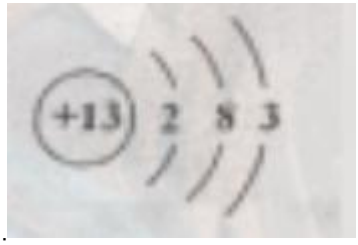
A.B<Al<Mg<Na　　B.Al<B<Mg<Na

C.Al<B<Na<Mg　　D.B<Al<Na<Mg

Analysis: For elements in the same period of the main group, the atomic radius decreases progressively from left to right (Na> Mg> Al); for elements in the same main group, the atomic radius increases progressively from top to bottom (Al> B). Therefore, the atomic radius order is (B <Al <Mg <Na).

33. The electron configuration diagram of the Na (sodium) element is ()

A. 

B. 

C 

D 

Analysis: A. Sodium (Na) has an atomic number of 11, with 11 protons in its nucleus and 11 electrons orbiting the nucleus.

At 34.25°C, the pH of 100mL of 0.05 mol·L $^{-1}$ H_2SO_4 solution is () after the water is evaporated to concentrate it to 10 mL.

A. 1

B. 2

C. 14

D. 0

Analysis: Solution Dilution and pH Calculation

Calculate the molar quantity of H^+in the original solution:

H_2SO_4 is a binary strong acid, c (H+) = 2 × 0.05 mol·L $^{-1}$= 0.1 mol·L $^{-1}$, n (H+) = 0.1 mol·L $^{-1}$ × 0.1 L = 0.01 mol.

After evaporating and concentrating to 10 mL (0.01 L), c(H^+) = $\frac{0.01\ \text{mol}}{0.01\ \text{L}} = 1\ \text{mol}\cdot\text{L}^{-1}$, pH = $-\lg c(H^+) = -\lg 1 = 0$。

35. In the reaction $2SO_2(g) + O_2(g) \rightarrow 2SO_3(g)$ in a 2L closed container, the amount of reactant SO_2 decreases from 1 mol to 0.6 mol within 10s. The reaction rate v(O2) is ().

A.0.02mol· $L^{-1} \cdot s^{-1}$

B.0.01mol· $L^{-1} \cdot s^{-1}$

C.0.2mol· $L^{-1} \cdot s^{-1}$

D.0.04mol· $L^{-1} \cdot s^{-1}$

Explanation: Calculation of Chemical Reaction Rate

Calculate v(s02) first:

$\Delta n(SO_2) = 1\ \text{mol} - 0.6\ \text{mol} = 0.4\ \text{mol}$,

$v(SO_2) = \frac{\Delta n}{V \cdot \Delta t} = \frac{0.4\ \text{mol}}{2\ \text{L} \times 10\ \text{s}} = 0.02\ \text{mol}\cdot\text{L}^{-1}\cdot\text{s}^{-1}$。

According to the reaction $2SO_2(g) + O_2(g) = 2SO_3(g)$, the rate ratio is equal to the stoichiometric ratio, i.e. v(02): v(s02) = 1: 2,

so $v(O_2) = \frac{1}{2}v(SO_2) = 0.01\ \text{mol}\cdot\text{L}^{-1}\cdot\text{s}^{-1}$。

36. The Reagent Used to Detect Cl‧ Ions Is ().

A.$AgNO_3$ and dilute sulfuric acid

B·$BA(N0)_2$ and dilute nitric acid

C.$BaCl_2$ and dilute hydrochloric acid

D.$AgNO_3$ and dilute sulfuric acid

Analysis: Principle of Chloride Ion Testing

The method to check cl-is to add $AgNO_3$ solution to produce white precipitate, and then add dilute nitric acid, the precipitate does not dissolve (eliminate $CO23^-$and other interference).

A: $SO24^-$in dilute sulfuric acid will form Ag_2SO_4 with $Ag^+$and interfere with the test;

B:$AgNO_3$ and dilute nitric acid, in accordance with the Cl-test method;

C:$BaCl_2$ itself contains cl-, which can cause interference;

D: Dilute sulfuric acid will introduce $SO24^-$and interfere with the test.

37 After the following reactions reach equilibrium, reducing pressure or increasing temperature can shift the chemical equilibrium toward the reaction direction. A.$N_2(g)$ +3 $H_2(g) \rightarrow 2NH_3(g)$ is an endothermic reaction. B.$2NO_2(g) \rightarrow N_2 + O_4$ is an exothermic reaction. Should

C.$3O_2(g) \rightarrow 2O_3$ The forward reaction is an endothermic reaction D.C (s) +$CO_2(g) \rightarrow$ 2 CO (g) forward and reverse

Analysis of

Endothermic Reaction: Principle of Chemical Equilibrium Shift (Le Chatelier's Principle)

Decreasing pressure shifts equilibrium toward the direction of increased gas volume; increasing temperature shifts equilibrium toward the direction of endothermic reaction. Both conditions must be met simultaneously:

A:$N_2(g)$ +$3H_2(g) \rightarrow 2NH_3(g)$ (endothermic reaction, gas volume decreases). Reducing pressure reverses the equilibrium, and increasing temperature reverses the equilibrium, which is inconsistent.

B:$2NO_2(g)$ is closed $N_2O_4(g)$ (exothermic reaction, gas volume decreases), reducing pressure equilibrium is reverse, increasing temperature equilibrium is reverse, does not conform;

C:$3O_2(g)$ The equilibrium is reversed when the pressure decreases and the temperature increases, which is inconsistent with the positive reaction endothermic and the gas volume decreases.

D: C (S) +$CO_2(g) \rightarrow$ 2C0 (g) (endothermic reaction, gas volume increases), decreasing pressure favors the forward equilibrium, increasing temperature favors the forward equilibrium, which is consistent with the principle.

38. Which of the following chemical reactions is a substitution reaction?

A. Reaction of ethylene with hydrogen chloride

B. Reactions of Bismuth with Hydrogen

C. Reaction of methane with chlorine D. Reaction of ethylene with water Analysis:

The reaction between ethylene and hydrogen chloride is an addition reaction, in which the carbon-carbon double bond in ethylene is opened and combines with HCl to form chloroethane.

The reaction of benzene with hydrogen is an addition reaction, in which the benzene ring adds to H_2 to form cyclohexane.

The reaction of methane with chlorine gas is a substitution reaction, in which the hydrogen atoms in methane are gradually replaced by chlorine atoms, forming products such as monochloromethane and dichloromethane.

The reaction of ethylene and water is an addition reaction, in which ethylene and water add together to form ethanol.

39. Which of the Following Statements About Organic Compounds Is Correct?

A. Organic compounds contain only carbon and oxygen elements. B. Organic compounds cannot be synthesized through artificial methods.

C. Organic compounds exhibit isomerism. D. Compounds containing carbon elements can be inorganic substances or

Therefore, it is organic matter.

Analysis: The compounds containing carbon elements can be inorganic or organic: Correct, such as CO,CO_2 belong to inorganic, methane, ethanol belong to organic.

40. When using solar energy to decompose water for hydrogen production, if 0.02mol of

water is decomposed, the following statement is correct.

A. The volume of H_2 generated is 0.224L (standard state). B. The amount of substance of H_2 generated is 0.01mol.C. The amount of substance of H_2 generated is 0.02 mol D.The mass of the generated H_2 is 0.02g

$$2H_2O \xrightarrow{\text{electrolyze}} 2H_2\uparrow + O_2\uparrow$$

Given the decomposition of 0.02mol H_2 0, according to the stoichiometric ratio:

n (H2) = n (H_2 0) = 0.02mol, the volume under standard conditions v = n × V_m= 0.02mol × 22.4L/mol = 0.448L;

$m(H_2) = n \times M = 0.02\text{mol} \times 2\text{g/mol} = 0.04\text{g}$;

$n(O_2) = \frac{1}{2}n(H_2O) = 0.01\text{mol}$

41 The molar concentration of the solution of the predatory solenium is 0.5 mol. L^{-1} is ()

A.1L contains a solution of 40g NaOH　　B.11.2 L HCl is dissolved in water to prepare a 1L solution

D.20g of NaOH dissolved in 1 L of water

C.1L solution of K2SO4 containing 1 mol of K+

Analysis: According to the molar concentration formula$c = \frac{n}{V}$ (v is the volume of the solution, in liters):

K_2SO_4中$n(K_2SO_4) = \frac{1}{2}n(K^+) = 0.5$ mol, $c = \frac{0.5\text{ mol}}{1\text{ L}} = 0.5\text{ mol} \cdot \text{L}^{-1}$, accord with ；

42. For the reaction C (S) +H_2 0 (g) ⟶ C0 (g) +H_2(g), the reaction equilibrium shifts in the reverse direction for ()

A. cooling and pressure reduction

B.Heating and pressurizing

C. cooling and blood pressure reduction

D.Cooling and pressurization

43. The liquid that forms a red Cu20 precipitate with newly prepared Cu(OH)2 is ()

A.CH_3COOH　　B.$CH_3COOCH_2CH_3$

C.CH_3CHO　　D.CH_3CH_2OH

Analysis: Characteristic reactions of the aldehyde group (silver mirror reaction/reaction with freshly prepared CU(OH)2)

The substance that can form red Cu_2 0 precipitate with newly prepared CU (0H)$_2$ must contain an aldehyde group (-CHO). Analyze the options one by one:

(A) CH_3 C00H: Acetic acid, containing carboxyl group (-C00H), reacts with freshly prepared CU(0H)$_2$ in neutralization without forming red precipitate;

(B) CH3C00CH2CH3: Ethyl acetate, containing an ester group, does not undergo the characteristic reaction with freshly prepared CU(0H)2;

(C) CH_3 CH0: Acetaldehyde, containing aldehyde group, can be heated with newly made CU (0H)2 to form red Cu_2 0 precipitation;

(D) CH_3CH_2 0H: ethanol, containing hydroxyl (-0H), does not react with newly prepared CU (0H)$_2$.

44. Which of the following ion groups can coexist in large quantities in the specified solution?

A. In alkaline solution: Al $^{3+}$,HCO_3^-, Cl-, K$^+$B. In acidic solution: Ag$^+$,NO_3^-, Cl-Na$^+$

C. In alkaline solution: Ba $^{2+}$, CO3$^-$, Cl-, K$^+$In acidic solution: Fe $^{3+}$,SO2$^-$,NO_3^-,Mg $^{2+}$

Analysis: There are a large number of H$^+$: Fe $^{3+}$, S024$^-$,NO_3^-, Mg $^{2+}$in the acidic solution, which do not react with each other, and can exist stably in the acidic environment, and can coexist in large quantities.

45. A mixture of CuO and Cu powder (23.4g) was reduced with CO. After complete reaction, 20.2g of pure copper powder was obtained. Thus, the mass of CuO in the original mixture is ().

A.20g B.32g

C.40g D.16g

Analysis: The chemical equation for CO reducing CuO is CU0 + C0$\overset{\Delta}{===}$CU +CO_2

The mass difference of the solid before and after the reaction is due to the oxygen element in CuO is taken away, the amount of the solid mass reduction is:

$$23.4\text{g} - 20.2\text{g} = 3.2\text{g}。$$

Assuming the mass of CuO in the original mixture is x, according to the chemical equation, CuO (relative molecular mass 80) reacts to form Cu (relative atomic mass 64). For every 80 parts of CuO that react, the solid mass decreases by 80-64 = 16 parts.

$$\frac{80}{16} = \frac{x}{3.2\text{g}}$$

The result is x = 16g.

46. In a chemistry lab session, the students performed the following procedures to achieve the objective: ()

Operation in experiment	Experimental objective
① Add excess NaOH solution to the $ZnCl_2$ solution	Preparation of zn(0H)$_2$ precipitate
② Heat and evaporate the $AlCl_3$ solution	Wastewater $AlCl_3$ prepared by $AlCl_3$ solution
③ Add dilute nitric acid solution, filter, wash, and dry	Remove the mixed CuO from the Cu content.
④ CO_2 is passed into the C_6H_5 ONa solution and the solution becomes turbid	Comparison of the acidity of carbonic acid and phenol

A.④ Experimental Procedures and Objectives B.① Experimental Procedures and Objectives

C.③ Experimental Procedures and Objectives　　D.② Experimental Procedures and Objectives

Analysis:CO_2 is introduced into the solution of C_6H_5 ONa, and the reaction occurs:

$$C_6H_5ONa + CO_2 + H_2O \rightarrow C_6H_5OH + NaHCO_3,$$

C_6H_5 0H is slightly soluble in water, making the solution turbid. According to the principle of "strong acid making weak acid", it can be proved that carbonic acid is stronger than phenol, and the purpose can be achieved.

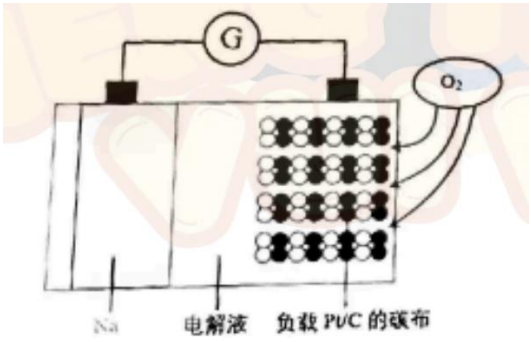
47 A white powder consists of two substances. To determine its composition, conduct the following experiments: ① Add a small sample to an adequate amount of water; the solid portion dissolves. Then add an adequate amount of dilute hydrochloric acid; bubbles form, and the solid dissolves completely. ② Add a small sample to an adequate amount of dilute sulfuric acid; bubbles form, and after shaking, solid remains. The white powder could be ().

A.$AgNO_3$,Na_2CO_3　　B.NaOH,$CUSO_4$

C.$CaCO_3$,$BaCO_3$　　D.Na_2CO_3,$BaCO_3$

Analysis: When water is added,Na_2CO_3 dissolves while $BaCO_3$ remains insoluble (solid part dissolves); when dilute hydrochloric acid is added,$BaCO_3$ reacts with HCl to form $BaCl_2$、CO_2 gas, with the solid completely dissolved; when dilute sulfuric acid is added,$BaCO_3$ reacts with H_2SO_4 to form $BaSO_4$ precipitate (insoluble in acid) and CO_2 gas, which matches the phenomenon of "bubbles present but solid still remaining," aligning with the experiment.

48. Scientists have developed a room-temperature Na-O_2 secondary battery, using $NaClO_4$ dissolved in an organic solvent as the electrolyte. The carbon cloth of Pt/C metal is used as the electrode material, and the overall reaction of the battery is:O_2+ 2Na → Na_2O_2. The following statement is correct



A. During discharge,ClO_4^-moves to the positive electrode　　B. It releases O_2 during discharge and absorbs O_2 during charging. C. During discharge, the negative electrode reaction is: Na^++ e- → Na D. During charging, the positive electrode reaction is:

$$Na_2O_2 - 2e^- \rightarrow 2Na^+ + O_2 \uparrow$$